SCHOOL OF MECHANICAL ENGINEERING (SMEC)

About School

The School of Mechanical Engineering is among the founding schools of VIT-AP UNIVERSITY at Amaravati that produce high quality engineers in the field of Mechanical Engineering, to cater to the demands of the nation and to move at par with the highly industrialized and technologically advanced modern world. The primary objective of the school is to impart knowledge-centered education and mould students into future engineers to meet global challenges in the area of Mechanical Engineering, thereby, enabling them to contribute to the societal prosperity of mankind. The School offers undergraduate programme (B.Tech.) with one of the most advanced and upto-date academic curricula including NASSCOM courses that are designed keeping in view the industrial requirements in India and overseas. The School aims to produce quality professionals in Mechanical Engineering to compete globally by carrying out the basics and applied research in emerging areas by forging strong industry–institute interaction. With the academic curriculum, the students are encouraged to be a part of various clubs for the overall personality development.

Programmes Offered

Program covers the entire spectrum of new-age specializations. This program seamlessly integrates basic science concepts to the latest technologies with relevant practice through lab and projects for building problem-solving skills

- Undergraduate

  B. Tech. Mechanical Engineering
  B. Tech. Mechanical AD
  B. Tech. Mechanical Robotics

- Doctoral

  Mechanical Engineering

UNDERGRADUATE - MECHANICAL-ENGINEERING

About the Programme

The undergraduate programme in Mechanical Engineering has a strong focus on the design of mechanical systems and hands-on experience. The programme includes in-depth study of thermal, design, manufacturing, and automation courses at core curriculum level to design the systems relevant for industries. In addition to the theoretical and laboratory based curriculum, students have to complete engineering clinics courses and a capstone project. The students have to complete a co-op in the industry for one semester. This degree programme provides a strong foundation in Mechanical Engineering discipline, development of critical thinking, and problem solving skills. Besides the technical skills, the academic programmes also sharpen communication, writing, and problem solving skills.

Eligibility

Nationality

- The applicant should be a Resident / Non-Resident Indian National / PIO / OCI.
- NRI applicants can directly apply under 'NRI Category' through the NRI application form.
- Foreign applicants who have studied/are studying abroad can apply directly through the International application form.
- Visit https://vit.ac.in/admissions/international/overview to know more.

Age Limit

- Applicants whose date of birth falls on or after 1st July 2002 are eligible to apply for admission into Engineering in the academic year 2024.

- The date of birth as recorded in the High School / SSC / X Certificate will be considered authentic.
- Applicants should produce this certificate in original as a proof of their age at the time of counselling/admission, failing which their candidacy for admission will be cancelled.

Fee Structure

INR 1,95,000 *

Tuition Fees

INR 3,000 *

Caution Deposit

(Refundable)

INR 1,98,000 *

Tuition Fees

(*Total fees to be paid for first year)

Career Prospects

- Mechanical Engineering graduates have huge employment opportunities in the automobile, manufacturing, instrumentation, and mechanical engineering design oriented industries.

MECHANICAL-ENGINEERING-AUTOMOTIVE-DESIGN

About the Programme

The automotive industry consistently seeks individuals with profound multidisciplinary expertise and a creative mindset to positively apply their knowledge. In today's context, the future of automotive vehicles lies in electrical and hybrid technologies. Moreover, the rise of predictive vehicle technology, fuel-efficient transportation, and self-driving capabilities represents sectors where emerging technologies present abundant opportunities. To delve into the intricacies of the automotive realm, specializing in Automotive Design is imperative.

Career Prospects

Students specializing or minoring in Automotive Design can explore promising career paths in core sectors such as the automobile industry, including renowned companies such as TATA, Hero, TVS, Hyundai, Volvo, and more. Those intrigued by the role of an Automotive Design Engineer can find fulfilling career opportunities. As the automotive landscape transitions towards electric vehicles, the establishment of manufacturing plants in India by industry giants such as Ola, Tesla, and Kia creates numerous openings. Additionally, students with an interest in software can pursue diverse career opportunities in this dynamic field.

SAME FEES AND ELIGIBILITY.

MECHANICAL-ENGINEERUNG-ROBOTICS

About the Programme

From the "Chandrayaan-I" initiative aimed at deploying robots on the moon to applications in biomedical engineering and the automotive sector, robots are playing a pivotal role across diverse fields. In a nation driven by technology, the significance of robotics has expanded beyond industrial use to encompass

various everyday human activities. Anticipate the transformative impact of robots as the next world-changing technology. Prepare to craft your robotic mastery – opt for a specialization in Robotics.

Career Prospects

Students specializing or minoring in robotics can explore promising career paths in advanced manufacturing units adopting Industry 4.0 standards. Opportunities include roles as a Robot Engineer. Given the multidisciplinary nature of robotics, additional avenues open up in the fields of Artificial Intelligence and Electronics.

SAME FEES AND ELIGIBILITY.

Ph.D. Program

Overview

Research is at the heart of SMEC's vision towards becoming an internationally recognized center of knowledge and learning. The spectrum of research activities at SMEC is extensive and distinct. Our researchers work to find out the technological solutions to real-world problems and to innovate new technologies to meet emerging world needs. Our labs have state-of-the-art facilities to carry projects for the development of technologies. Currently, the school has over 37 Research Scholars pursuing a Ph.D. degree. The faculty members have contributed to academic research publishing and have successfully published 89 research papers in internationally renowned referred journals and several book chapters. SMEC encourages our researchers for collaborative research with faculties from foreign universities and presents their findings on international platforms. The school is regularly hosting international conferences to create an international conglomeration of scientists, engineers, and industrial experts. The focus of the conferences is to provide a technical platform that encourages scientific research and educational activities that would cater to the needs of both society and the industry. The school has about 7 registered faculties under the Research Grant in Engineering, Management, and Science (RGEMS), an initiative to motivate research that can lead to R&D projects. Undergraduate Research (URE) pursuits are encouraged by the school as well. In collaboration with the faculty, undergraduate students pursue active interdisciplinary research to make intellectual or creative contribution towards advancing Science. SMEC Highlights include Ongoing 4 External Funded Projects, All faculty members of SMEC are from reputed institutions across the globe, MoU's with reputed academic, research institutes and industries, About 20 Indian Patents publications by faculty and students of SMEC, and About 89 research publications by faculty and students of SMEC.

Major Areas of Research

Polymer nanocomposites, Honeycomb sandwich structures, Biomedical materials, Auxetic materials, Materials Processing, Polymer Engineering, Machining Science, Metal additive manufacturing, Finite Element Analysis, Structural Health Monitoring, Tribology, Numerical Modelling of Manufacturing Processes, Metal Forming, Railway Engineering, Optical diagnostics, Image processing, Experimental and numerical investigations of droplet and spray processes. Thermal Engineering, Refrigeration & Air Conditioning, Heat Transfer, Heat Exchanges,

Fees

Programme	Ph.D. Internal Full Time
Tuition Fees (INR)	40,000
Caution Deposit (INR)	3,000
Total Fees (INR)	43,000
Thesis Fee (INR)	10,000